

Zstate Pilot Status Brief - July 2026 (WIP)

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Work in progress - 2026-07-02 git a638bc0

Internal team share. Covers architecture, pilot tasks, reward design, PM simulator, and early model findings. Not for external publication.

1. What we are building

Zstate is an equity-research benchmark with two complementary tracks in one repo: Track A (public leaderboard, single-turn forensics/modeling tasks) and Track B (RL training env with a skeptical Portfolio Manager). Both share fixed EDGAR corpora, expert ground truth, and a fracture taxonomy.

Architecture (high level)

Shared foundation: Corpus (fixed doc bundles) + Expert pipeline (CFA draft/review) + Trajectory schema v1

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+-- Track A (benchmark_v0.1/) -- single-turn Type F/M tasks
|   -> 3-layer reward (L1 accuracy / L2 gold path / L3 citations)
|   -> Leaderboard + fracture codes
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+-- Track B (env_v1/) -- dual-control episodes
|   -> PM policy FSM (pushback, follow-ups, pushover trap)
|   -> 4-component reward (Outcome / Grounding / Defense / Hallucination)
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+-- Track C (future export/) -- curated JSONL + reward vectors for training product
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2. Reward systems

Track A - 3-layer benchmark score

Layer | What it measures | Typical weight (Type F)

L1 Hard accuracy | Python verify vs expert ground truth (metrics, arithmetic) | 55%

L2 Gold path | Section recall, tool order, workflow fidelity | 25%

L3 Trust / citations | Verbatim snippets, policy acks, anti-hallucination | 20%

Composite = weighted sum. Fracture codes tag failure modes (e.g. CITE_BROAD).

Track B - 4-component env reward

Reward = 0.45 Outcome + 0.25 Grounding + 0.20 Defense - 0.10 Hallucination. Defense is env-only: rewards substantive engagement when the PM pushes back. Capitulation without retrieval triggers pushover penalties.

PM simulator (Track B)

Scripted FSM (pm_v1_1): opening skepticism -> follow-up A/B/C branches -> pushover trap.

Agent tools: fixed corpus search + send_message_to_pm + submit_recommendation.

Sample episode: Solaris earnings-quality dispute (env_v1/episodes/).

Defense is the new discriminator vs single-turn QA - see frontier runs in env_v1/runs/.

3. Published pilot tasks (5/15 MVD)

Task ID | Ticker | Type | Tier | Objective

GOOGL_footnote_reconciliation | GOOGL | F | L2 | Using Alphabet Inc.'s Form 10-Q for the quarter ended March 31, 2026 (filed April 30, 2026), reconcile reportable seg...

PEP_fx_organic_growth | PEP | M | L3 | Using PepsiCo, Inc.'s Form 10-K for the fiscal year ended December 27, 2025 (filed February 3, 2026), compute constan...

AMZN_footnote_reconciliation | AMZN | F | L3 | Using Amazon.com, Inc.'s Form 10-K for the fiscal year ended December 31, 2025 (filed February 6, 2026), reconcile re...

NFLX_guidance_drift | NFLX | F | L3 | Using Netflix, Inc. materials for FY2025 guidance and Q3 2025 actuals, quantify guidance drift on **content cash

spen...

KO_footnote_reconciliation | KO | F | L3 | Using The Coca-Cola Company's Form 10-K for the fiscal year ended December 31, 2025 (filed February 20, 2026), reconc...

Task highlights

GOOGL (F): Q1 2026 segment sum + hedging reconciling item - ceiling task for frontier models.

PEP (M): FY2025 FX / organic growth decomposition from MD&A tables.

AMZN (F): FY2025 segment operating income bridge with stock-based comp decoy.

NFLX (F): FY2025 cash content guidance vs nine-month YTD actuals (guidance drift).

KO (F, new): FY2025 five-segment Total net revenues + Corporate + Eliminations bridge to \$47,941M consolidated; LatAm (2)% total vs (12)% FX from MD&A.

4. Eval methodology

Campaign runner: benchmark_v0.1/scripts/run_benchmark_campaign.py

Eval mode ON: generic citation rules only (no task-specific cheat-sheets).

Path roles: canonical slugs (segment_financials, narrative_fx) - not issuer note numbers.

Headline composite excludes GOOGL (ceiling); PEP + AMZN + NFLX (+ KO in 5-task campaigns).

Models routed: OpenAI (gpt-*), Anthropic (claude-*), Gemini (gemini-* via OpenAI-compatible API).

3 runs per task (median) for all models in pilot_eval_5task_v1.

5. Findings (July 2026 pilot)

Full 5-task eval (pilot_eval_5task_v1)

Campaign pilot_eval_5task_v1: 45 runs scored (5 tasks x 3 models x 3 runs). Headline = PEP + AMZN + NFLX + KO (excl. GOOGL).

Model | Headline weighted composite

claude-sonnet-4-5 | 0.994

gemini-2.5-flash | 0.982

gpt-4o | 0.896

Per-task medians by model:

Task | claude sonnet 4 5 | gemini 2.5 flash | gpt 4o

AMZN footnote reconciliation | 1.000 | 1.000 | 0.958

GOOGL footnote reconciliation | 1.000 | 1.000 | 1.000

KO footnote reconciliation | 1.000 | 1.000 | 0.884

NFLX guidance drift | 0.976 | 0.927 | 0.879

PEP fx organic growth | 1.000 | 1.000 | 0.864

Fractures (all runs): {'CITE_HALLUC': 6, 'CITE_BROAD': 16}

Interpretation:

KO_footnote_reconciliation is the universal gap - all three models median 0.0 on headline.

Claude leads headline (0.744); Gemini close second (0.732); GPT-4o third (0.675).

GOOGL ceiling task: all models median 1.0 on 3 runs.

L3 citation fractures (CITE_BROAD, CITE_HALLUC) dominate; KO adds HALLUC_FILL on L1.

6. WIP / next steps

Scale to 15-task MVD (5 companies x 3 task types) with expert sign-off pipeline.

Investigate KO metric schema confusion (wrong submit keys observed on Gemini).

Track B: more frontier trajectories + PM branch coverage.

Expert Workbench (spec) for CFA draft -> review -> publish workflow.

LATER-06: full corpus ingest with excerpt SHA pins (P3-10 mitigation on KO bundle).

References: docs/ARCHITECTURE.md

benchmark_v0.1/docs/PILOT_EVAL_JUL2026.md

env_v1/docs/METHODOLOGY_RL_ENV.md